

Q. 2(i)

force₁ = 9N

force₂ = 12N

$$\begin{aligned}\text{magnitude} &= \sqrt{(9)^2 + (12)^2} \\ &= \sqrt{81 + 144} \\ &= \sqrt{225} \\ &= 15\text{m}\end{aligned}$$

(b) direction :-

opposite : 12N , adjacent : 9N

$$\tan \theta = \frac{\text{opposite}}{\text{adjacent}}$$

$$\tan \theta = \frac{12}{9}$$

$$\theta = \tan^{-1}\left(\frac{4}{3}\right)$$

$\theta \approx 53.1^\circ$ from the 9N force.

Q. 2(ii)

(a)

Clockwise

moment

=

Anticlockwise

moment

$$W_1 \times d_1 = W_2 \times d_2$$

$$6 \times 20 = W_2 \times 15\text{cm}$$

$$120 \text{ Nm} = W_2 \times 15 \text{ cm}$$

$$\frac{120 \text{ Nm}}{15 \text{ cm}} = W_2$$

$$8 \text{ N} = W_2$$

(b)

Principle of moments.

Q.2(iii)

(a)

$$m = 50 \text{ kg}$$

$$h = 4 \text{ m}$$

$$g = 10 \text{ m/s}^2$$

$$t = 8 \text{ s}$$

$$W = mg$$

$$= 50 \times 10$$

$$500 \text{ N}$$

$$\text{Work} = mgh$$

$$= 50 \times 10 \times 8$$

$$= 2000 \text{ J}$$

(b) $P = \frac{\text{Work}}{\text{Time}}$

$$\frac{2000}{8} = 250 \text{ W}$$

Q. 3(i)

(a)

$$m = 600 \text{ kg}$$

$$v_i = 0 \text{ m/s}$$

$$v_f = 20 \text{ m/s}$$

$$t = 10 \text{ s}$$

$$\vec{a} = \frac{v_f - v_i}{t}$$

$$= \frac{20 - 0}{10}$$

$$= \frac{20}{10} \Rightarrow 2 \text{ m/s}^2$$

(b) Distance :- ?

$$s = v_i \times t + \frac{1}{2} a t^2$$

$$= 0 \times 10 + \frac{1}{2} (2) (10)^2$$

$$= 0 + (10)^2$$

$$s = 100 \text{ m}$$

(c)

$$F = m a$$

$$= 600 \times 2$$

$$F = 1200 \text{ N}$$

$$(d) \quad p = mv$$

change in momentum =

$$\Delta p = mv_f - mv_i$$

$$= 600 \times 20 - 600 \times 0$$

$$= 12000 - 0$$

$$\Delta p = 12000 \text{ Kg} \cdot \text{m/s}$$

Q: 3(ii)

$$F = 100 \text{ N}$$

$$A_1 = 0.004 \text{ m}^2$$

$$A_2 = 0.2 \text{ m}^2$$

(a)

$$P = \frac{F_1}{A_1}$$

$$= \frac{100}{0.004}$$

$$= 25000 \text{ Pa}$$

$$= 25000 \text{ Pa}$$

(b)

$$P = \frac{F_2}{A_2}$$

$$F_2 = P \times A_2$$

$$= 25000 \times 0.2$$

$$F_2 = 5000 \text{ N}$$

(c)

~~$$F_2 = P \times A_2$$~~

$$\frac{F_2}{F_1} = \frac{A_2}{A_1}$$

$$\frac{5000}{100} = \frac{0.2}{0.004}$$

$$50 = 50$$

(d)

A liquid is used because liquids are nearly incompressible, so the pressure applied to the ~~small~~ small piston is transmitted effectively to the large piston.

A gas is compressible, so it would compress instead of transmitting the pressure effectively.

MCQs.

1. A

$$v_i = 5 \text{ m/s}$$

$$v_f = 20 \text{ m/s}$$

$$t = 5 \text{ s}$$

$$a = \frac{v_f - v_i}{t}$$

$$= \frac{20 - 5}{5}$$

$$= \frac{15}{5} \Rightarrow 3 \text{ m/s}^2$$

2. B

$$p = mv$$

$$= 4 \times 7$$

$$= 28 \text{ kg} \cdot \text{m/s}$$

3. D

Reason :- action and reactions are forces b/w two interacting object. we cannot do action and reaction on the same body. when bodies are touching or on bodies that are at rest.

4. C

reason: other options have different definitions.

5. A

reason: the condition clearly said that it must be zero so other options aren't true.

6. C

$$P = \rho gh$$

$$= 1000 \times 10 \times 4$$
$$40,000 \text{ Pa}$$

7. B

$$f = 20 \text{ N}$$

$$d = 0.3$$

$$W = f \times d$$
$$20 \times 0.3$$
$$= 6 \text{ N}$$

8. D

$$\text{power} = \text{Work} \div \text{time}$$
$$1800 \div 6$$
$$= 300 \text{ W}$$

9. A

reason: other options aren't true.

10. B
reason:- because the ~~ans~~ statement
said diagnosis, treatment,
respiration so it connects
to biology.